PRISMATIC COHOMOLOGY VAR TC^- AND NYGAARD FILTRATIONS

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Abstract. This report is for the advanced course on prismatic cohomology at AMSS during 2025 fall. In this report, We will introduce the construction of Nygaard-complete prismatic cohomology var topological cyclic world and compare the Nygaard filtration with Hahn-Raksit-Wilson even filtration. We will prove a generalized version of comparison theorem and give some applications in q-de Rham cohomology. More explicitly, we will finish the construction of q-Hodge filtration on q-de Rham complex by explaining a comparison theorem of Ferdinand Wagner, which also replenishes the author's talks on the same topic at MCM in this seminar.

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Introduction

The prismatic cohomology serves as a package in the arithmetic world. The center object of the prismatic theory is the animated ring $\mathbb{\Delta}_{R/A}$ that computes the cohomology, which is associated with crystalline cohomology, étale cohomology, q-de Rham cohomology, negative cyclic homology and more. It is therefore of great interest to ask how to reflect the prismatic cohomology from the others. The purpose of this report is to explain the relationship between the prismatic cohomology and negative cyclic homology. More explicitly, we will prove the following theorem.

Theorem 0.1. *Let $S \in \mathrm{QSyn}$. We have an equivalence of filtrations.*

$$\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S)^\wedge_p \simeq \mathcal{N}^{\geqslant i}\hat{\mathbb{\Delta}}_S$$

Here on the left side of the equivalence "HRW" means Hahn-Raksit-Wilson even filtration, which we will introduce in Section 1.4. The right side of the equivalence is the Nygaard filtration, which we will define in Section 2.

The ultimate goal of this report is to present a detailed proof of the following theorem of [[Wag]; A.17], which applys the above consequence immediately and connects prismatic cohomology with the q-de Rham cohomology.

Theorem 0.2 ([Wag]). *Let S be a quasisyntomic $A[\zeta_p]$-algebra in p-complete category, then there is an equivalence of graded $\mathbb{E}_\infty$-$\mathbb{Z}_p[\![q-1]\!][u,v]/(uv-[p]_q)$-algebras*

$$\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S/\mathbb{S}_A[\![q-1]\!])^\wedge_p \simeq \mathcal{N}^{\geqslant *}\hat{\mathbb{\Delta}}^{(p)}_{S/A[\![q-1]\!]}$$

This theorem shows how to reflect the prismatic information from negative cyclic homology. Moreover, with a bit more efforts, one can show that after inverting u on each side and taking the 0-th graded pieces, one has

$$\mathrm{gr}^0_{\mathrm{HRW}}\mathrm{TC}^-(S/\mathbb{S}_A[\![q-1]\!][u^{-1}])^\wedge_{p,q-1} \simeq \mathbb{\Delta}_{S/A[\![q-1]\!]}[u^{\pm 1}]$$

Combining with [[BS22]; Theorem 16.18], which states that $q\text{-dR}_{R/A} \simeq \mathbb{\Delta}_{R^{(p)}[\zeta_p]/A[\![q-1]\!]}$, one has

Corollary 0.3. *We have an equivalence*

$$q\text{-dR}_{R/A} \simeq \mathrm{gr}^0_{\mathrm{HRW}}\mathrm{TC}^-(R^{(p)}[\zeta_p]/\mathbb{S}_A[\![q-1]\!][u^{-1}])^\wedge_{p,q-1}$$

This allows us to construct more structures over q-de Rham complexes. For example, one can construct the q-Hodge filtration var pullback along the above equivalence, as shown in [[Wag]; Section 7].

In order to complete the proof, we will divide the contents as follows. In the first part we begin with some recollection of basic facts. We will quickly review some facts about topological Hochschild homology and negative cyclic homology in Section 1.2, which is one of the basic object we are dealing with through this report. We concentrate in a special range of rings, which we refer as quasisyntomic rings and define them in Section 1.3. We briefly introduce the Hahn-Raksit-Wilson even filtrations and show they satisfy specific descent.

The whole contents of Section 2 is devoted to define Nygaard filtration on various objects (i.e. TC^-, A_{inf} and $\mathbb{\Delta}_S$) and compare them with HRW-even filtration. In this section, we will prove our first main theorem Theorem 0.1. The most challenging part of the proof is to describe THH, TC^- and TP of quasi syntomic rings. In order to do this, we will first discuss the structure of $\mathrm{TC}^-(R)$ in Section 2.1 for all perfectoid rings R. Afterwards, we can unpack the information of quasisyntomic rings as algebras over perfectoid rings using the tricks of [NS18]. We then briefly explain the relationship between prismatic cohomology and TC^- in Section 2.3.

We will apply this comparision theorem in q-de Rham cohomology in Section 3. In order to do this, we need to generalize the consequences of [BMS19] and [Wag] to the relative case

over an $\mathbb{E}_\infty$-base. We state and prove this generalized theorem in Section 3.1, and complete the proof of Theorem 0.2 in Section 3.2.

The main reference for this report is [BMS19].

1. Setting the stage

The purpose of this section is to explain the notations and set the stage for the discussion in the next section.

1.1. **Notations and prerequisites.** Throughout this report, we will feel free to use the ∞-categorical language. Without particular specification, algebras here mean that of the ∞-categorical context with at least $\mathbb{E}_1$-structures and all rings are $\mathbb{E}_\infty$-rings.

We say that a spectrum R is **even** if the (stable) homotopy group of R concentrates in even degrees.

The filtrations in this report mean **descending filtrations**. That is, filtrations of the form

$$\mathrm{fil}^* X = \Big(\cdots \leftarrow \mathrm{fil}^i X \leftarrow \mathrm{fil}^{i+1} \leftarrow \cdots \Big)$$

We let $\mathrm{gr}^* X$ be the associated graded pieces, which is defined as $\mathrm{gr}^n X := \mathrm{cofib}(\mathrm{fil}^{n+1} \to \mathrm{fil}^n)$. We let Σ^{2*} be the **double sheering** functor, which is defined as

$$\Sigma^{2*} : \mathrm{Gr}(\mathcal{C}) \to \mathrm{Gr}(\mathcal{C})$$

$$\{\mathrm{gr}^n(X)\} \mapsto \{\mathrm{gr}^n(X)[2n]\}$$

The double shearing functor is $\mathbb{E}_1$-monoidal, as shown in [Rak20].

Throughout the report, we fix a prime p and concentrate on p-complete category. Therefore, all the operators, such as THH, TC$^-$, TP and tensors are in the p-complete sense. We will write down the p-completion as far as possible. But sometimes we will omit this for notation simplicity.

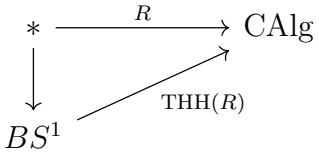
1.2. **Topological Hochschild homology.** We now introduce the topological Hochschild homology. The main reference for this part is [NS18]. We begin with the definition, which is placed in the ∞-categorical settings.

Definition 1.1. Let R be an $\mathbb{E}_1$-ring spectrum, we define the **topological Hochschild homology** of R to be

$$\mathrm{THH}(R) = R \otimes_{R \otimes R^{op}} R$$

The ring spectrum $\mathrm{THH}(R)$ admits a natural S^1-action. To see this, we need to notice that $S^1 \simeq * \sqcup_{* \sqcup *} *$. In fact, the following theorem shows that the functor $\mathrm{THH}(X)$ is somehow a "free" S^1 equivariant spectrum generated by X.

Theorem 1.2 (McClure-Schwänzel-Vogt)**.** *The functor* $\mathrm{THH} : \mathrm{CAlg} \to \mathrm{CAlg}^{BS^1}$ *is the left Kan extension of the constant map.*



Therefore, one has an equivalence

$$\mathrm{Map}_{\mathrm{CAlg}}(R, -) \simeq \mathrm{Map}_{\mathrm{CAlg}^{BS^1}}(\mathrm{THH}(R), -)$$

The computation of THH is not a long story. The first computable example is given by Bökstedt.

Theorem 1.3 ([Bök85])**.** *There is an equivalence of* $\mathbb{E}_1$*-rings*

$$\mathrm{THH}(\mathbb{F}_p) \simeq \mathbb{F}_p[\Omega S^3]$$

Therefore, we have $\mathrm{THH}(\mathbb{F}_p)_* \simeq \mathbb{F}_p[u]$ *with the generator* u *in degree* 2*.*

We are more interested in the relative version. Note that there is a canonical map $\mathrm{THH}(A) \to A$. Therefore, for any $\mathbb{E}_1$-A-algebra R one can define

$$\mathrm{THH}(R/A) = \mathrm{THH}(R) \otimes_{\mathrm{THH}(A)} A$$

When $A = \mathbb{S}$, this coincides with the above definition; when $A = \mathbb{Z}$, this coincides with the traditional Hochschild homology. The relative THH also admits an S^1-action. Therefore, one can talk about the homotopy fixed points and Tate-constructions.

Definition 1.4. We define the negative cyclic homology as

$$\mathrm{TC}^-(R/A) := \mathrm{THH}(R/A)^{hS^1}$$

We define the periodic cyclic homology as

$$\mathrm{TP}(R/A) := \mathrm{THH}(R/A)^{tS^1}$$

Proposition 1.5. *We have the base change formula for map of* $\mathbb{E}_\infty$*-maps* $A \to R$*,* $A \to S$ *and* $S \to B$*.*

$$\mathrm{THH}(R/A) \otimes_S B \simeq \mathrm{THH}(R \otimes_S B / A \otimes_S B)$$

Remark 1.6. After base change to $\mathbb{Z}$, we obtain the classical definition of HH. Recall that by Hochschild–Kostant–Rosenberg theorem, if A is smooth over R, then we have an equivalence $\Omega^*_{A/R} \simeq \mathrm{HH}(A/R)$. This endows $\mathrm{HH}(A/R)$ with a filtration, which we refer to as **HKR-filtration**. The graded pieces of this filtration is given by $\mathrm{gr}^i_{\mathrm{HKR}}\mathrm{HH}(-/R) \simeq \wedge^i L_{-/R}[i]$.

We conclude this subsection by the following theorem, which is a generalization of [[BMS19]; Corollary 3.4]

Theorem 1.7. *Let $\mathbb{S}_A$ be an $\mathbb{E}_\infty$-ring with $\mathbb{S}_A \otimes_{\mathbb{S}} \mathbb{Z} = A$ for some ordinary ring A, then the functors $\mathrm{THH}(-/\mathbb{S}_A)$, $\mathrm{TC}^-(-/\mathbb{S}_A)$ and $\mathrm{TP}(-/\mathbb{S}_A)$ are spectra-value sheaves on CRings with respect to the fpqc topology.*

Proof. It suffices to check the sheaf condition for $\mathrm{THH}(-/\mathbb{S}_A)$ and $\mathrm{THH}(-/\mathbb{S}_A)_{hS^1}$, then the sheaf condition of TC^- follows because it is a limit and that of TP follows because of the Tate cofiber sequence. For $\mathrm{THH}(-/\mathbb{S}_A)$, note that $\mathrm{THH}(-/\mathbb{S}_A) \otimes_{\mathrm{THH}(\mathbb{Z})} \otimes \tau_{\leqslant n}\mathrm{THH}(\mathbb{Z})$ is a Postnikov tower with the limit $\mathrm{THH}(-/\mathbb{S}_A)$. By induction, it suffices to check the sheaf condition for $\mathrm{THH}(-/\mathbb{S}_A) \otimes_{\mathrm{THH}(\mathbb{Z})} \otimes \pi_{\leqslant n}\mathrm{THH}(\mathbb{Z})$. By finiteness of $\mathrm{THH}(\mathbb{Z})$, it reduces to check $\mathrm{THH}(-/\mathbb{S}_A) \otimes_{\mathrm{THH}(\mathbb{Z})} \otimes \pi_{\leqslant n}\mathrm{THH}(\mathbb{Z})$, which by base change and our requirement for $\mathbb{S}_A$ is equivalent to $\mathrm{HH}(-/A)$. It follows from [[BMS19]; Corollary 3.4] that $\mathrm{HH}(-/A)$ is a spectra-value sheaf.

For $\mathrm{THH}(-/\mathbb{S}_A)_{hS^1}$, the procedure is similar. We need only to apply homotopy orbit functor to the above process. □

1.3. **Quasisyntomic site.** We are interested in a special class of rings, which we call them **quasisyntomic**. In this section we present the definition of QSyn and some properties of the quasisyntomic site. We recall the definition of Tor-amplitude.

Definition 1.8. Let R be a ring. Given $a, b \in \mathbb{Z}$ and $M \in D(R)$, we say M has **Tor-amplitude** in $[a, b]$ if for any R-module N, $M \otimes^{\mathbb{L}}_R N \in D^{[a,b]}(R)$. We say $M \in D(R)$ has **p-complete Tor-amplitude** in $[a, b]$ if $M \otimes^{\mathbb{L}}_R R/p \in D(R/p)$ has Tor-amplitude in $[a, b]$. We say M is **p-completely (faithfully) flat** if $M \otimes^{\mathbb{L}}_R R/p \in D(R/p)$ concentrates in degree 0 and is (faithfully) flat.

Definition 1.9 (Quasisyntomic site)**.** We define the quasisyntomic site through the following steps.

(1) A ring A is **quasisyntomic** if A is p-complete with bounded p^∞-torsion and $L_{A/\mathbb{Z}_p} \in D(A)$ has p-complete Tor-amplitude in $[-1, 0]$.
(2) A map $f : A \to B$ is **quasismooth** if B is p-completely flat over A and $L_{B/A} \in D(B)$ is p-completely flat.
We say a quasismooth map is a quasismooth cover if it is faithfully flat.
(3) A map $f : A \to B$ is **quasisyntomic** if B is p-completely flat over A and $L_{B/A} \in D(B)$ has p-complete Tor-amplitude in $[-1, 0]$.
We say a quasisyntomic map is a quasisyntomic cover if it is faithfully flat.

Quasisyntomic rings and quasisyntomic maps form a category QSyn. We equip it with quasisyntomic covers.

Remark 1.10. The condition $L_{R/A}$ has Tor-amplitude in $[-1, 0]$ is sometimes referred to as **quasi-lci**, see for example [Wag].

There are many interesting examples of quasisyntomic rings. For example, $\mathbb{Z}_p[\zeta_p]$, which is of vital importance in the construction of q-Hodge filtrations.

Our first aim of this subsection is to prove that QSyn equipped with quasisyntomic covers indeed form a site. We begin with the following lemma.

Lemma 1.11 ([BMS19]; Lemma 4.7)**.** *Let A is an ordinary ring with bounded p^∞-torsion.*

(1) *Suppose $M \in D(A)$ is derived p-complete and p-completely flat, then M is a classically p-complete A-module (i.e. concentrate in degree 0) with bounded p^∞-torsion, such that M/p^nM is plat over A/p^nA for $n \geqslant 1$. Moreover, for any $n \geqslant 1$, we have an isomorphism*

$$M \otimes_A A[p^n] \xrightarrow{\cong} M[p^n]$$

(2) *Conversely, suppose N is classically p-complete A-module with bounded p^∞-torsion such that N/p^nN is flat over A/p^nA for any $n \geqslant 1$, then $N \in D(A)$ is p-completely flat.*

The following corollary follows immediately.

Corollary 1.12. *Let $A \to B$ be a map of derived p-complete rings. If A has bounded p^∞-torsion and $A \to B$ is p-completely flat, then B has bounded p^∞-torsion.*

We are now finish the preparation that we need in order to prove that QSyn^{op} indeed forms a site.

Theorem 1.13. *The quasisyntomic covers satisfy the followings.*

(1) *Isomorphisms are quasisyntomic covers.*
(2) *The composision of quasisyntomic covers is a quasisyntomic cover.*
(3) *In the category of p-complete rings, suppose $A \to C$ is a quasisyntomic cover and $A \to B$ is an arbituary map where A, B, C are all quasisyntomic, then the pushout $D = B \otimes^\wedge_A C$ is quasisyntomic and $B \to D$ is a quasisyntomic cover.*

Proof. The first assertion is trivial. For the stability under composition, we consider the transitivity triangle of cotangent complexes associated with the composition $A \to B \to C$

$$L_{B/A} \otimes^{\mathbb{L}} C \longrightarrow L_{C/A} \longrightarrow L_{C/B}$$

Since B is quasi-lci over A and C is quasi-lci over B, C must be quasi-lci over A. Moreover, $A \to C$ is faithfully flat by the stability of faithfully flatness under composition.

We now prove the last claim. Let D' be the derived p-completion of $B \otimes^{\mathbb{L}}_A C$. By base change, D' is faithfully flat over B. Therefore, by Lemma 1.11, $D \simeq D'$. Since faithfully flatness and the formation of cotangent complexes are both stable under base change, we conclude that $B \to D$ is a quasisyntomic cover.

It remains to show that D is quasisyntomic. We consider the transitivity triangle of cotangent complexes associated with the composition $\mathbb{Z}_p \to B \to D$

$$L_{B/\mathbb{Z}_p} \otimes^{\mathbb{L}} D \longrightarrow L_{D/\mathbb{Z}_p} \longrightarrow L_{D/B}$$

Therefore one can conclude that D is quasi-lci over B, hence quasisyntomic. □

Remark 1.14. The above theorem holds true if we replace every "quasisyntomic cover" by "quasismooth cover". Indeed, in this case we need only to check the cotangent complexes concentrate in degree 0 instead of $[-1, 0]$, which can be done similarly by examining the transitivity triangles.

Remark 1.15. It is clear that the quasisyntomic topology is slightly coarser than fpqc topology. Therefore, by Theorem 1.7, the cyclic world players THH, TC$^-$ and TP define spectra-value sheaves on QSyn^{op} as well.

In order to survey the TC$^-$ on quasisyntomic rings for later use, we will need a new basis for QSyn, which we defined as **quasiregular semiperfectoid rings**. Recall that a ring R is called **perfectoid** if it is p-complete, R/p is semiperfect, there exist $\pi \in R$ and $u \in R^\times$ such that $\pi^p = pu$ and the kernel of $\theta : \mathrm{A_{inf}}(R) \to R$ is generated by one element. The perfectoid rings are all quasisyntomic.

Definition 1.16. A ring S is quasiregular semiperfectoid if

(1) S is quasisyntomic.
(2) There exists some $R \to S$ with R perfectoid.
(3) S/p is semiperfect.

We use QRSPerfd to denote the category of all quasiregular semiperfectoid rings and equip it with quasisyntomic covers.

Our next aim is to show that under its cover QRSPerfd^{op} forms a site and compare it with QSyn^{op}.

Proposition 1.17. *The category* QRSPerfd^{op} *equipped with quasisyntomic covers forms a site.*

Proof. The only none-trivial assertion is the existence of the pushout. Suppose $A \to C$ is in QRSPerfd and $A \to B$ is arbituary, then we have shown that $B \to C \otimes^{\wedge}_{A} B$ is a quasisyntomic cover. The bounded torsion part and surjection of Frobenius then follow from Lemma 1.11. □

We now compare QRSPerfd^{op} with QSyn^{op}. We will need the following lemma.

Lemma 1.18. *If a p-complete ring A admits a quasisyntomic cover $A \to S$ with $S \in$* QRSPerfd^{op}*, then $A \in$* QSyn*. Moreover, in this case, the Čech nerve $S^\bullet$ lies in* QRSPerfd^{op}*.*

Therefore, by taking the totalization of Čech nerve and other formal arguments, one can construct inverse functors between the sheaf categories of these two sites with values in any presentable categories.

Corollary 1.19. *For any presentable ∞-category $\mathcal{C}$, there is an equivalence*

$$\mathrm{Shv}_{\mathcal{C}}(\mathrm{QSyn}^{op}) \simeq \mathrm{Shv}_{\mathcal{C}}(\mathrm{QRSPerfd}^{op})$$

where the inverse equivalence is obtained by totalizing the quasiregular semiperfectoid resolution as defined in above lemma.

1.4. **HRW-even filtrations.** Filtrations allow us to convert analyzation of spectra to graded pieces, which will expose more information. For example, one can often analyze THH by the associated HKR-filtration. In this subsection, we will define the Hahn-Raksit-Wilson even filtration and discuss the descent property. The main reference for this subsection is [HRW22].

Proposition 1.20. *Let* CAlg^{ev} *denote the full subcategory of* CAlg *of even $\mathbb{E}_\infty$-rings.* CAlg^{ev} *is an accessible subcategory of* CAlg*.*

Proof. We can consider the homotopy group functor, which fits into the following pullback diagram.

$$\begin{array}{ccc} \mathrm{CAlg}^{ev} & \longrightarrow & \mathrm{CAlg} \\ {\scriptstyle \pi_*}\downarrow & \lrcorner & \downarrow{\scriptstyle \pi_*} \\ \mathrm{GrAb}^{ev} & \longrightarrow & \mathrm{GrAb} \end{array}$$

Both the right arrow and the bottom arrow are continuous. Therefore, by [[Lur09]; Proposition 5.4.6.6], CAlg^{ev} is an accessible subcategory of CAlg. □

We have a double speed Whitehead filtration functor for even spectra, which takes an even spectra X to $(\cdots \to \tau_{\geqslant 2n}X \to \tau_{\geqslant 2n-2}X \to \cdots)$. Moreover, this functor admits a natural lax symmetric monoidal structure.

Definition 1.21. We define the **HRW-even filtration** $\mathrm{fil}^*_{\mathrm{ev}} : \mathrm{CAlg} \to \mathrm{FilCAlg}$ to be the right Kan extension

$$\begin{array}{ccc} \mathrm{CAlg}^{ev} & \hookrightarrow & \mathrm{CAlg} \\ & {\scriptstyle \tau_{\geqslant 2*}}\searrow & \downarrow{\scriptstyle \mathrm{fil}^*_{\mathrm{ev}}} \\ & & \mathrm{FilCAlg} \end{array}$$

More explicitly, we have the following expression:

$$\mathrm{fil}^*_{\mathrm{ev}}(A) = \lim_{A \to B, B \in \mathrm{CAlg}^{ev}} \tau_{\geqslant 2*}(B)$$

Remark 1.22. There are some variants of this construction. First, if we focus on p-complete world, we will obtain an even filtration functor $\mathrm{fil}^*_{p,\mathrm{ev}} : \mathrm{CAlg}_p \to \mathrm{FilCAlg}_p$ by a similar right Kan extension. Moreover, one can similarly define $\mathrm{fil}^*_{p,hS^1,\mathrm{ev}}$ and $\mathrm{fil}^*_{p,tS^1,\mathrm{ev}}$ from $\mathrm{CAlg}_p^{BS^1}$ to $\mathrm{FilCAlg}_p$ by right Kan extension of $(\tau_{\geqslant 2*}(-))^{hS^1}$ and $(\tau_{\geqslant 2*}(-))^{tS^1}$. Finally, if we are interested in cyclotomic structures, we can also define $\mathrm{fil}^*_{p,\mathrm{TC},\mathrm{ev}} : \mathrm{CycCAlg}_p \to \mathrm{FilCAlg}_p$ in a similar method.

Remark 1.23. We need to be careful in the p-complete world. The author thanks Jeremy Hahn for explaining this to me and any flaw in this remark is my own mistake. When defining $\mathrm{fil}^*_{p,\mathrm{ev}} : \mathrm{CAlg}_p \to \mathrm{FilCAlg}_p$, the full subcategory CAlg_p^{ev} of CAlg_p should be chosen to be the full subcategory of p-complete even $\mathbb{E}_\infty$-algebras of bounded p^∞-torsion instead of simply choosing p-complete even $\mathbb{E}_\infty$-algebras. That means, we require the homotopy group of an object in CAlg_p^{ev} to be of bounded p^∞-torsion. The bounded torsion requirement here is to make sure that after inverting one element in p-complete category the object is still even. See [[BL22]; Section 1.7] for an animated version of explanation.

We now discuss the descent property of HRW-even filtrations. We first define topologies on various even subcategories.

Definition 1.24. Let $f : A \to B$ be a morphism in CAlg_p^{ev}. We say f is p-completely faithfully flat if the associated map of graded homotopy rings is p-completely faithfully flat.

Proposition 1.25. *The pushout of p-completely faithfully flat morphism along any map is p-completely faithfully flat.*

Proof. Since everything is even, then the Tor-spectral sequence computing the homotopy group of tensor products degenerate. The proposition follows immediately from the stability of p-completely faithfully flat morphism along pushout of derived rings. □

We can equip $(\mathrm{CAlg}_p^{ev})^{op}$ with the p-complete flat topology. That is, a collection of morphisms form a covering if and only if it contains a finite collection of maps $\{A \to B_i\}$ with $A \to \prod_i B_i$ p-completely faithfully flat.

Lemma 1.26. $(\mathrm{CAlg}_p^{ev})^{op}$ *equipped with the p-complete flat covering topology forms a site.*

Proof. This follows from [[Lur18]; A.3.2.1] and Proposition 1.25. □

Remark 1.27. Since pushouts in $\mathrm{CAlg}_p^{BS^1}$ is computed termwise in CAlg_p^{ev}, $\mathrm{CAlg}_p^{BS^1}$ can be equipped with the induced topology, which we also refer to as p-complete flat topology.

We want to use this flat descent to compute the homotopy group. Therefore, we need the following consequence.

Proposition 1.28. *(1) The functor $\pi_{2*} : \mathrm{CAlg}_p^{ev} \to \mathrm{GrRing}$ is a sheaf with respect to the p-complete flat topology.*
(2) The functor $\pi_{2*}(-)^{hS^1}$ and $\pi_{2*}(-)^{tS^1} : (\mathrm{CAlg}_p^{ev})^{BS^1} \to \mathrm{GrRing}$ is a sheaf with respect to the p-complete flat topology.

We need the following lemma in order to prove the above proposition.

Lemma 1.29. *Let A be an even $\mathbb{E}_\infty$-ring with S^1-action. Then*
(1) There is a class $t \in \pi_{-2}(A^{hS^1})$ which is a nonzerodivisor in $\pi_*(A^{hS^1})$ and the map $\pi_*(A^{hS^1}) \to \pi_*(A^{tS^1})$ is given by inverting t.
(2) For any S^1-equivariant A-module M, the canonical maps

$$M^{hS^1} \otimes_{A^{hS^1}} A \to M \quad M^{hS^1} \otimes_{A^{hS^1}} A^{tS^1} \to M^{tS^1}$$

are equivalences. M^{hS^1} is t-complete.

Proof of Proposition 1.28. For the first assertion, let $A \to B$ be a p-completely faithfully flat map in CAlg_p^{ev}. We need to show that

$$\pi_{2*}(A) \to \lim_{\Delta} \pi_{2*}((B^{\otimes_A \bullet})_p^\wedge) \simeq \lim_{\Delta} (\pi_{2*}(B)^{\otimes_{\pi_{2*}(A)} \bullet})_p^\wedge$$

is an equivalence. This comes from the p-complete version of faithfully flat descent of [[Lur18] ; Theorem D.6.3.5].

We now prove the second one. Let $A \to B$ be a p-completely faithfully flat map in $(\mathrm{CAlg}_p^{ev})^{BS^1}$. We need to show that there are equivalences

$$\pi_{2*}(A^{hS^1}) \xrightarrow{\simeq} \lim_{\Delta} \pi_{2*}((B^{\otimes_A \bullet})_p^\wedge)^{hS^1} \quad \pi_{2*}(A^{tS^1}) \xrightarrow{\simeq} \lim_{\Delta} \pi_{2*}((B^{\otimes_A \bullet})_p^\wedge)^{tS^1}$$

Choose $t \in \pi_{-2}(A^{hS^1})$ which is a nonzerodivisor in $\pi_*(A^{hS^1})$. For homotopy fixed points case, by Lemma 1.29, both sides are t-complete so it suffices to prove the claim after taking the

derived base change along $\mathbb{Z}_p[t] \to \mathbb{Z}_p$. This recovers the map $\pi_{2*}(A) \to \lim_\Delta \pi_{2*}((B^{\otimes_A \bullet})^\wedge_p)$, which is an equivalence by the first assertion. The Tate construction case is obtained by inverting t on both sides for the homotopy fixed points case, which holds true because taking the totalization commutes with inverting t in the category of rings. □

Corollary 1.30. *We focus on p-complete categories.*

1. *The functor $\pi_{\geqslant 2*} : \mathrm{CAlg}_p^{ev} \to \mathrm{FilCAlg}$ is a sheaf with respect to the p-complete flat topology.*
2. *The functor $\pi_{\geqslant 2*}(-)^{hS^1}$ and $\pi_{\geqslant 2*}(-)^{tS^1} : (\mathrm{CAlg}_p^{ev})^{BS^1} \to \mathrm{FilCAlg}$ is a sheaf with respect to the p-complete flat topology.*

Proof. In the even case all the filtrations are complete, so it suffices to check the sheaf condition on graded pieces, which follows immediately from Proposition 1.28. □

We can state the descent for general $\mathbb{E}_\infty$-algebras, which is our main tool to contral the even filtrations. We will need to define a new class of morphisms, which we will see in the last section to cover a large class of morphisms.

Definition 1.31 (evenly faithfully flat)**.** We say a map $f : A \to B$ is p-**complete eff**(evenly faithfully flat) if for any $A \to C$ with $C \in \mathrm{CAlg}_p^{ev}$, $(B \otimes_A C)^\wedge_p$ is even and p-completely faithfully flat. We say a S^1-equivariant map $f : A \to B$ is p-complete eff if its underlying map of $\mathbb{E}_\infty$-algebras is p-complete eff.

Combining with Corollary 1.30 and base change, one obtains immediately the following consequence.

Corollary 1.32 (Eff descent)**.** *We focus on p-complete world.*

1. *For $f : A \to B$ a p-complete eff map in CAlg_p, we have an equivalence*

$$\mathrm{fil}^*_{p,\mathrm{ev}}(A) \to \lim_\Delta \mathrm{fil}^*_{p,\mathrm{ev}}((B^{\otimes_A \bullet})^\wedge_p)$$

2. *For $f : A \to B$ a p-complete eff map in $(\mathrm{CAlg}_p)^{BS^1}$, we have equivalences*

$$\mathrm{fil}^*_{p,hS^1,\mathrm{ev}}(A) \to \lim_\Delta \mathrm{fil}^*_{p,hS^1,\mathrm{ev}}((B^{\otimes_A \bullet})^\wedge_p) \quad \mathrm{fil}^*_{p,tS^1,\mathrm{ev}}(A) \to \lim_\Delta \mathrm{fil}^*_{p,tS^1,\mathrm{ev}}((B^{\otimes_A \bullet})^\wedge_p)$$

2. Nygaard filtrations

The main goal of this section is to introduce the Nygaard filtration on TC^- and compare it with HRW-even filtration. In this section we will prove our first main theorem. We will begin with some structure consequences of THH of quasisyntomic rings, which will "incubate" a variety of filtrations on it.

2.1. THH of perfectoid rings.

Since we focus on the algebras over perfectoid rings, we begin with the computation of perfectoid rings. Through this section, we fix such a perfectoid ring R and $A_{\mathrm{inf}} = \mathrm{A_{inf}}(R)$, with $\theta = \theta_R : A_{\mathrm{inf}} \to R$.

The most fundamental example on our hands is $R = \mathbb{F}_p$. Recall that by Theorem 1.3, $\pi_*\mathrm{THH}(\mathbb{F}_p) \cong \mathbb{F}_p[u]$ with u a single generator in degree 2. This inspires and anticipates the following theorem.

Theorem 2.1. *Let R be a p-complete perfectoid ring. There is an isomorphism*

$$\pi_*\mathrm{THH}(R)^\wedge_p \cong R[u]$$

with u a single generator in degree 2. Moreover, u is a generator of $\pi_2(\mathrm{THH}(R)^\wedge_p) \cong \pi_2(\mathrm{HH}(R)^\wedge_p) \cong \ker(\theta)/\ker(\theta)^2$.

Proof. By HKR filtration, the graded pieces of $\mathrm{HH}(R)^\wedge_p$ are given by $(\wedge^i_R L_{R/\mathbb{Z}_p})^\wedge_p[i] \simeq R[2i]$, we have $\pi_i\mathrm{HH}(R)^\wedge_p \cong R$ if $i \geq 0$ is even and $= 0$ else. Therefore, $\mathrm{HH}(R)^\wedge_p$ is a pseudocoherent complex of R-modules. Since we have

$$\mathrm{THH}(R)^\wedge_p \otimes_{\mathrm{THH}(\mathbb{Z})} \mathbb{Z} = \mathrm{HH}(R)^\wedge_p$$

By induction, all $\mathrm{THH}(R)^\wedge_p \otimes_{\mathrm{THH}(\mathbb{Z})} \tau_{\leqslant n}\mathrm{THH}(\mathbb{Z})$ are pseudocoherent, which implies that $\mathrm{THH}(R)^\wedge_p$ itself is pseudocoherent.

We need to check that for any map $R \to S$ between perfectoid rings the induced map

$$\mathrm{THH}(R)^\wedge_p \otimes_R S \to \mathrm{THH}(S)^\wedge_p$$

is an equivalence. By base change formula and induction, we need only to check the assertion after base change to $\mathbb{Z}$. That means, we need only to check

$$\mathrm{HH}(R)^\wedge_p \otimes_R S \to \mathrm{HH}(S)^\wedge_p$$

is an equivalence, which is again by the HKR filtration.

The theorem holds true for $R = \mathbb{F}_p$ by Theorem 1.3. Thus, the above equivalence implies that it holds if R is of characteristic p.

The general case follows by induction on i. We may suppose result holds true in degrees $< i$. Note that $\tau_{\leqslant i-1}\mathrm{THH}(R)^\wedge_p$ is a perfect complex of R-modules, it follows that $\tau_{\geqslant i}\mathrm{THH}(R)^\wedge_p$ is still pseudocoherent. Therefore, $M = \pi_i\mathrm{THH}(R)^\wedge_p$ is a finitely generated R-module. Consider the map

$$M' = \begin{cases} R \cdot u^{i/2} & i \text{ even} \geq 0, \\ 0 & \text{else} \end{cases} \to M = \pi_i\mathrm{THH}(R)^\wedge_p$$

and let $\overline{R}$ be the direct limit perfection of R/p. Then $R \to \overline{R}$ is surjective and the kernel lies in the Jacobson radical, $\overline{R}$ is a perfect ring of characteristic p. By the base change formula and freeness of $\pi_j\mathrm{THH}(R)^\wedge_p$ for $j < i$, we see that

$$M \otimes_R \overline{R} = \pi_i\mathrm{THH}(\overline{R}),$$

which by the case of characteristic p is given by $M' \otimes_R \overline{R} = \overline{R} \cdot u^{i/2}$ in even degrees. Thus, the map $M' \otimes_R \overline{R} \to M \otimes_R \overline{R}$ is an isomorphism.

Finally, if i is odd, then $M \otimes_R \overline{R} = 0$, which by Nakayama's lemma implies that $M = 0$ as desired. If i is even, then Nakayama's lemma implies that $M' \cong R \to M$ is surjective. To see that it is an isomorphism, it suffices to see that the rank of M at all points of $\mathrm{Spec}R$ is at least 1 as R is reduced. This can be checked after base change to $\mathbb{Q}$. □

We now use the above structure theorem of THH to calculate the other players, i.e. TC$^-$ and TP. Recall that by [[NS18]; Lemma II.4.2], we have the following square, where φ is the cyclotomic Frobenius.

$$\begin{array}{ccc} \mathrm{TC}^-(R) & \xrightarrow{\varphi^{hS^1}} & \mathrm{TP}(R) \\ \downarrow & & \downarrow \\ \mathrm{THH}(R) & \xrightarrow{\varphi} & \mathrm{THH}(R)^{tC_p} \end{array}$$

Proposition 2.2. *We have equivalences*

$$\mathrm{TC}^-(R)^\wedge_p \simeq A_{\mathrm{inf}}[u,v]/(uv-\xi) \quad \mathrm{TP}(R)^\wedge_p \simeq A_{\mathrm{inf}}[\sigma,\sigma^{-1}] \quad (\mathrm{THH}(R)^\wedge_p)^{tC_p} \simeq R[\sigma,\sigma^{-1}]$$

where ξ has degree 0 and is a generator of $\ker(\theta)/\ker(\theta)^2$; u and σ are in degree 2; v is in degree -2. Moreover, apply the above square, we obtain

$$\begin{array}{ccc} A_{\mathrm{inf}}[u,v]/(uv-\xi) & \xrightarrow{\varphi^{hS^1}} & A_{\mathrm{inf}}[\sigma,\sigma^{-1}] \\ \downarrow & & \downarrow \\ R[u] & \xrightarrow{\varphi} & R[\sigma,\sigma^{-1}] \end{array}$$

where the upper arrow takes u to σ and v to $\xi\sigma^{-1}$.

Proof. We first consider $\pi_0\mathrm{TP}(R)^\wedge_p$, which can be computed by the Tate spectral sequence. As a result, $\mathrm{TP}(R)^\wedge_p$ is concentrated in even degrees, and $\pi_0\mathrm{TP}(R)^\wedge_p$ admits a complete descending filtration $\mathrm{fil}^*\pi_0\mathrm{TP}(R)^\wedge_p$ with $\mathrm{gr}^i\pi_0\mathrm{TP}(R)^\wedge_p \cong \pi_{2i}\mathrm{THH}(R)^\wedge_p \cong R$ in even degrees. In particular, $\pi_0\mathrm{TP}(R)^\wedge_p \to \pi_0\mathrm{THH}(R)^\wedge_p = R$ is a p-complete pro-nilpotent thickening. By the universal property of A_{inf} [[Fon94]; Section 1.2], there is a unique map $A_{\mathrm{inf}} \to \pi_0\mathrm{TP}(R)^\wedge_p$ which sends $\ker(\theta)$ into $\mathrm{fil}^1\pi_0\mathrm{TP}(R)^\wedge_p$, and thus by multiplicativity, this sends $\ker(\theta)^i$ into $\mathrm{fil}^i\pi_0\mathrm{TP}(R)^\wedge_p$. We claim that this map between graded rings is infact a graded isomorphism. By completeness, it suffices to check on graded pieces. This can be checked after base change to $\mathbb{F}_p$, which follows from [[NS18]; Corollary IV.4.8]. Therefore by periodicity, there is an isomorphism $\pi_*\mathrm{TP}(R)^\wedge_p \cong A_{\mathrm{inf}}[\sigma,\sigma^{-1}]$ by choosing a generator $\sigma \in \pi_2\mathrm{TP}(R)^\wedge_p$.

The assertion for $\mathrm{TC}^-(R)^\wedge_p$ can be concluded similarly by analyzing the homotopy fixed points spectral sequence. That means we can find an isomorphism

$$\pi_*\mathrm{TC}^-(R)^\wedge_p \cong A_{\mathrm{inf}}[u,v]/(uv-\xi) \subset A_{\mathrm{inf}}[\sigma^{\pm 1}]$$

under which $v \mapsto \sigma^{-1}$ and $u \mapsto \xi\sigma$ under the canonical map.

We now identify the map $\pi_*(\varphi^{hS^1})$. It suffices to check on generators. First, that $\pi_2(\varphi^{hS^1})$ sends the generator to the generator can be checked after base change to $\mathbb{F}_p$, which is again by [[NS18]; Corollary IV.4.9]. In order to identify $\pi_0(\varphi^{hS^1})$, we note that there is a commutative square

$$\begin{array}{ccccc} \pi_0\mathrm{TC}^-(R)^\wedge_p & \xrightarrow{\pi_0(\varphi^{hS^1})} & \pi_0\mathrm{TP}(R)^\wedge_p & \longrightarrow & \pi_0(R^{tS^1}) \\ \downarrow & & \downarrow & & \downarrow \\ \pi_0\mathrm{THH}(R)^\wedge_p & \longrightarrow & \pi_0(\mathrm{THH}(R)^\wedge_p)^{tC_p} & \longrightarrow & \pi_0(R^{tC_p}) \end{array}$$

where the bottom horizontal composition is the Frobenius $\varphi : R \to R/p$. Therefore, $\pi_0(\varphi^{hS^1})$ fits into the square

$$\begin{array}{ccc} A_{\mathrm{inf}} & \xrightarrow{\pi_0(\varphi^{hS^1})} & A_{\mathrm{inf}} \\ {\scriptstyle\theta}\downarrow & & \downarrow{\scriptstyle\theta} \\ R & \xrightarrow{\varphi} & R/p \end{array}$$

By universal property of A_{inf}, the upper map must be the Frobenius.

Finally, we need to identify $(\mathrm{THH}(R)^\wedge_p)^{tC_p}$. By above discussion, it suffices to show that

$$(\mathrm{TP}(R) \otimes_{\mathrm{TC}^-(R)} \mathrm{THH}(R))^\wedge_p \to (\mathrm{THH}(R)^\wedge_p)^{tC_p}$$

is an equivalence. This follows from Lemma 1.29. □

We can now write down some structure consequences of quasisyntomic R-algebras. In fact, we have the following general property.

Proposition 2.3. *For any $\mathbb{E}_\infty$-R algebra A, we have following natural equivalences*

$$\mathrm{TC}^-(A)^\wedge_p/v \simeq \mathrm{THH}(A)^\wedge_p \quad \mathrm{TP}(A)^\wedge_p/\xi \simeq (\mathrm{THH}(A)^\wedge_p)^{tC_p}$$

Proof. This follows analogously from Lemma 1.29. □

2.2. **Nygaard filtrations on** TC$^-$**.** Before we define the Nygaard filtrations on TC$^-$, we recall the classical definition of Nygaard filtration on A_{inf}, which serves as a motivation of our definition of Nygaard filtration on TC$^-$ and TP.

Definition 2.4. We define the **Nygaard filtration** on A_{inf}, denoted by $\mathcal{N}^{\geqslant *}A_{\mathrm{inf}} := \mathrm{fil}^* A_{\mathrm{inf}}$, as

$$\mathcal{N}^{\geqslant *}A_{\mathrm{inf}} = \begin{cases} \xi^i A_{\mathrm{inf}} & \text{for } i \geqslant 0 \\ A_{\mathrm{inf}} & \text{otherwise} \end{cases}$$

It is clear that the i-th graded piece of this filtration is $\mathcal{N}^i A_{\mathrm{inf}} := \mathrm{gr}^i A_{\mathrm{inf}} = R\xi^i$

Let R be a p-complete perfectoid ring. By the computation in the above section, we have the following obvious equivalences.

$$\pi_*\mathrm{THH}(R)^\wedge_p \cong \bigoplus_{i\in\mathbb{N}} \mathcal{N}^i A_{\mathrm{inf}}\{i\}$$
$$\pi_*\mathrm{TC}^-(R)^\wedge_p \cong \bigoplus_{i\in\mathbb{Z}} \mathcal{N}^{\geqslant i} A_{\mathrm{inf}}\{i\}$$

$$\pi_*\mathrm{TP}(R)^\wedge_p \cong \bigoplus_{i\in\mathbb{Z}} A_{\mathrm{inf}}\{i\}$$

We are now going to move from perfectoid rings to quasiregular semiperfectoid rings.

Theorem 2.5. *Let R be a p-complete perfectoid ring and $S \in \mathrm{QRSPerfd}_R$.*

(1) $\mathrm{THH}(S)^\wedge_p$ is even.

(2) Multiplication by $u \in \pi_2\mathrm{THH}(R)^\wedge_p$ gives an injective map

$$\pi_{2i-2}(\mathrm{THH}(S)^\wedge_p) \xrightarrow{u} \pi_{2i}(\mathrm{THH}(S)^\wedge_p)$$

In order to prove this theorem, we need some preparation first.

Lemma 2.6 ([BMS19]; Theorem 6.7)**.** *Let A be a perfectoid ring. For any ordinary R-algebra A, there is an S^1-equivariant cofiber sequence*

$$\mathrm{THH}(A)^\wedge_p[2] \xrightarrow{u} \mathrm{THH}(A)^\wedge_p \to \mathrm{HH}(A/R)^\wedge_p$$

Therefore, by passing to homotopy fixed points and Tate constructions, there are cofiber sequences

$$\mathrm{TC}^-(A)^\wedge_p[2] \xrightarrow{u} \mathrm{TC}^-(A)^\wedge_p \to \mathrm{HC}^-(A/R)^\wedge_p$$

$$\mathrm{TP}(A)^\wedge_p[2] \xrightarrow{\xi\sigma} \mathrm{TP}(A)^\wedge_p \to \mathrm{HP}(A/R)^\wedge_p$$

Proof. We consider the sequence

$$\mathrm{THH}(R)^\wedge_p[2] \xrightarrow{u} \mathrm{THH}(R)^\wedge_p \to \mathrm{HH}(R/R)^\wedge_p$$

After applying $\mathrm{THH}(A)\otimes^\wedge_{\mathrm{THH}(R)}(-)$, one obtains the first sequence in the lemma. Therefore, it suffices to prove the lemma for $A = R$. However, the first cofiber sequence follows immediately by the structure of $\mathrm{THH}(R)$ as shown in Theorem 2.1 and the fact that $\mathrm{HH}(R/R) = R$. The second and third cofiber sequences follow from the exactness of homotopy fixed points functor and Tate construction. □

Using this lemma and some general facts about left Kan-extension, one can eventually obtain the following technical lemma.

Lemma 2.7. *The functor $\mathrm{THH}(-)^\wedge_p$ on the category of p-complete R-algebras admits a complete descending $\mathbb{N}$-indexed filtration $P^*(-)$ with gr^n_P identified with*

$$\bigoplus_{\substack{0\leqslant i\leqslant n\\ i-n\ even}} (\wedge^i L_{-/R})^\wedge_p[n]$$

We can now prove the theorem.

Proof of Theorem 2.5. By Lemma 2.7, the spectrum $\mathrm{THH}(S)^\wedge_p$ admits a complete filtration with

$$\mathrm{gr}^n \mathrm{THH}(S)^\wedge_p \cong \bigoplus_{\substack{0\leqslant i\leqslant n \\ i-n \text{ even}}} (\wedge^i L_{-/R})^\wedge_p[n]$$

Note that S is quasiregular semiperfectoid over R, the cotangent complex $\wedge^i L_{-/R})^\wedge_p$ concentrates in degree i by [[BMS19]; Lemma 5.14]. Therefore, the graded pieces are even, which implies $\mathrm{THH}(S)^\wedge_p$ is even by completeness.

The second claim then follows from Lemma 2.6. The cofiber sequence

$$\mathrm{THH}(S)^\wedge_p[2] \xrightarrow{u} \mathrm{THH}(S)^\wedge_p \to \mathrm{HH}(S/R)^\wedge_p$$

induces a short exact sequence

$$0 \to \pi_{2i-2}\mathrm{THH}(S)^\wedge_p \xrightarrow{u} \pi_{2i}\mathrm{THH}(S)^\wedge_p \to \pi_{2i}\mathrm{HH}(S/R)^\wedge_p \to 0$$

□

Now we can pass everything to the homotopy fixed points and Tate constructions.

Theorem 2.8. *Let $S \in \mathrm{QRSPerfd}_R$.*

1. *The homotopy fixed point spectral sequence calculating $\mathrm{TC}^-(S)^\wedge_p$ and the Tate spectral sequence calculating $\mathrm{TP}(S)^\wedge_p$ degenarate. As a consequence, both $\mathrm{TC}^-(S)^\wedge_p$ and $\mathrm{TP}(S)^\wedge_p$ are even. Moreover, the canonical map $\mathrm{TC}^-(S)^\wedge_p \to \mathrm{TP}(S)^\wedge_p$ induces injective maps on homotopy groups in each degree.*
2. *The degenerate spectral sequences endow $\hat{\mathbb{\Delta}}_S := \pi_0(\mathrm{TC}^-(S)^\wedge_p) \cong \pi_0(\mathrm{TP}(S)^\wedge_p)$ with the same complete descending $\mathbb{N}$-index filtration, called **Nygaard filtration**, which is denoted by $\mathcal{N}^{\geqslant *}\hat{\mathbb{\Delta}}_S := \mathrm{fil}^* \hat{\mathbb{\Delta}}_S$.*
3. *The filtration $\mathcal{N}^{\geqslant i}\hat{\mathbb{\Delta}}_S \subset \hat{\mathbb{\Delta}}_S$ is identified with $\pi_{2i}\mathrm{TC}^-(S)^\wedge_p$ var multiplication by $v^i \in \pi_{-2i}\mathrm{TC}^-(R)^\wedge_p$. The associated graded piece $\mathcal{N}^i\hat{\mathbb{\Delta}}_S$ is identified with $\pi_{2i}\mathrm{THH}(S)^\wedge_p$ for $i \geqslant 0$.*
4. *As a consequence, we have an equivalence of filtrations*

$$\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S)^\wedge_p \simeq \mathcal{N}^{\geqslant i}\hat{\mathbb{\Delta}}_S$$

5. *The cyclotomic Frobenius $\pi_*\mathrm{TC}^-(S)^\wedge_p \to \pi_*\mathrm{TP}(S)^\wedge_p$ induces a divided Frobenius $\varphi_S : \hat{\mathbb{\Delta}}_S \to \hat{\mathbb{\Delta}}_S$.*

Proof. The assertion (1) and (2) follow immediately, since now everything is even. The claim (3) is by examing the homotopy fixed points spectral sequence. (4) is given by (3) and the definition of double sheering in Section 1.1. Finally, the assertion (5) is obtained by the description of φ^{hS^1} from Proposition 2.2. □

By the equivalence of ∞-topoi of Corollary 1.19, the Nygaard filtration of TC$^-$ of quasiregular semiperfectoid rings naturally induces a filtration on TC$^-$ of quasisyntomic rings, which we also denoted by $\mathcal{N}^{\geqslant i}\hat{\mathbb{\Delta}}_S$. Our next aim is to show that this Nygaard filtration is still equivalent to the double sheering filtration of HRW-even filtration in this case.

Corollary 2.9. *Let $S \in \mathrm{QSyn}$. We have an equivalence of filtrations.*

$$\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S)^\wedge_p \simeq \mathcal{N}^{\geqslant i}\hat{\mathbb{\Delta}}_S$$

Proof. Note that both sides are completed filtrations. Therefore, it suffices to check equivalences on graded pieces. By Theorem 2.8 and quasisyntomic descent, $\mathrm{TC}^-(S)^\wedge_p$ is even so the graded pieces of HRW-even filtrations are single homotopy groups. Which by descent equivalent to the right side. □

2.3. **Prismatic cohomology var Nygaard completion.** In the previous subsection we introduce the notation $\hat{\mathbb{\Delta}}_S$. In this subsection we explain the relationship between $\hat{\mathbb{\Delta}}_S$ and prismatic cohomology theory. The main aim of this subsection is to state that $\hat{\mathbb{\Delta}}_S$ can be identified as the Nygaard completion of $\mathbb{\Delta}_S$ and the proof will be skimed.

Let us begin with some discussion of prisms and prismatic cohomology. We will concentrate in local cases. Suppose (A, I) is a bounded prism and X is a p-complete formal scheme over A/I. Recall that by Hodge-Tate comparison, we denote

$$\mathbb{\Delta}_{X/A} := R\nu_*\mathcal{O}_{\mathbb{\Delta}} \in D(X_{\acute{e}t}, A) \quad \text{and} \quad \overline{\mathbb{\Delta}}_{X/A} := R\nu_*\overline{\mathcal{O}}_{\mathbb{\Delta}} \in D(X_{\acute{e}t}, \mathcal{O}_X)$$

In the local situation, we define $\mathbb{\Delta}_{R/A} = R\Gamma(\mathrm{Spec}(R), \mathbb{\Delta}_{\mathrm{Spec}(R)/A})$ and $\overline{\mathbb{\Delta}}_{R/A} = \mathbb{\Delta}_{R/A} \otimes^{\mathbb{L}} A/I$. The ring $\mathbb{\Delta}_{R/A}$ is always static(i.e. discrete) if we are dealing with quasisyntomic rings, but in general $\overline{\mathbb{\Delta}}_{R/A}$ is merely an animated ring. However, we will see that it is indeed static if we focus on quasiregular semiperfectoid rings.

Proposition 2.10. *Let $S \in \mathrm{QRSPerfd}$. The category of prisms (A, I) equipped with a map $S \to A/I$ admits an initial object $(\mathbb{\Delta}^{\mathrm{ini}}_S, I)$.*

We need the following proposition, which makes it clear for us to define $\mathbb{\Delta}_S$.

Proposition 2.11 ([BS22]; Proposition 7.10)**.** *Let R be a perfectoid ring and $S \in \mathrm{QRSPerfd}_R$. We define $(A, I) := (\mathrm{A}_{\mathrm{inf}}(R), \ker(\theta))$.*

(1) $\overline{\mathbb{\Delta}}_{S/A}$ is static.
(2) $\mathbb{\Delta}_{S/A} \simeq \mathbb{\Delta}^{\mathrm{ini}}_S$.
(3) The ideal of $\mathbb{\Delta}^{\mathrm{ini}}_S$ is generated by a single generator.

In particular, $\mathbb{\Delta}_{S/A} \simeq \mathbb{\Delta}^{\mathrm{ini}}_S$ doesn't rely on R and we denote it by $\mathbb{\Delta}_S$.

We can define the Nygaard filtration on $\mathbb{\Delta}_S$ in a formal way.

Definition 2.12. Let S be a quasiregular semiperfectoid ring with associated prism $(\mathbb{\Delta}_S, (d))$. The Nygaard filtration on $\mathbb{\Delta}_S$ is given by

$$\mathcal{N}^{\geqslant i}\mathbb{\Delta}_S := \mathrm{fil}^i\mathbb{\Delta}_S = \{x \in \mathbb{\Delta}_S | \phi(x) \in d^i\mathbb{\Delta}_S\}$$

Theorem 2.13 ([BS22]; Theorem 13.1)**.** *There is a functorial δ-ring structure on $\hat{\mathbb{\Delta}}_S = \pi_0\mathrm{TC}^-(S)^\wedge_p = \pi_0\mathrm{TP}(S)^\wedge_p$ refining the cyclotomic endomorphism ϕ. The induced map $\mathbb{\Delta}_S = \mathbb{\Delta}^{\mathrm{ini}}_S \to \hat{\mathbb{\Delta}}_S$ identifies $\hat{\mathbb{\Delta}}_S$ with the Nygaard completion of $\mathbb{\Delta}_S$.*

In order to generalize the above notation to the relative settings, we will need the following definition, which defines a new basis for QSyn^{op}.

Definition 2.14. Let (A, I) be a bounded prism. A quasisyntomic A/I-algebra S is **large** if there is a surjection $A/I\langle X_i^{1/p^\infty} | i \in I\rangle \to S$ for some I.

Note that large quasisyntomic algebras form a basis for quasisyntomic site as one can always extract compatible sequences of p-power roots of elements.

We can now generalize the above notation of Nygaard filtration on the relative case.

Theorem 2.15. *Let (A, I) be a bounded prism. Let S be a large quasisyntomic A/I-algebra. Then the derived prismatic cohomology $\mathbb{\Delta}_{S/A}$ is a (p, I)-completely flat δ-algebra. It is the initial object of $(S/A)_{\mathbb{\Delta}}$.*

Therefore, let $\mathbb{\Delta}^{(1)}_{S/A} = \mathbb{\Delta}_{S/A} \otimes^{\mathbb{L}} A$ to be the Frobenius twist. One can define the Nygaard filtration on the relative derived prismatic cohomology after Frobenius twist by

$$\mathcal{N}^{\geqslant i}\mathbb{\Delta}^{(1)}_{S/A} := \mathrm{fil}^i\mathbb{\Delta}^{(1)}_{S/A} = \{x \in \mathbb{\Delta}^{(1)}_{S/A} | \phi(x) \in I^i\mathbb{\Delta}^{(1)}_{S/A}\}$$

In the following section, we will define the Nygaard filtration on TC$^-$ in relative case for a wide range of objects and prove a relative version of Theorem 2.13.

3. Application to q-de Rham cohomology

In this part, we will prove our second main theorem Theorem 0.2, which associates the negative cyclic homology and prismatic cohomology with q-de Rham cohomology. The author will improve the comparison theorems of Bhatt-Morrow-Scholze and Wagner, which will ultimately pack our proof in a very concise fashion.

We will prove this generalized version of theorem in Section 3.1. Then we will apply this theorem immediately in to practice in Section 3.2.

3.1. A generalization of the base change theorem of Bhatt-Morrow-Scholze.

The aim of this subsection is to prove a relative version of Corollary 2.9, which is a generalization of [[BMS19] ; Proposition 11.2], [[BS22] ; Proposition 15.7] and [[Wag]; Proposition A.17].

We make some requirements for the notations of this subsection

Notations 3.1. We let A be a p-complete and p-completely covered δ-ring with an $\mathbb{E}_\infty$-spherical lift $\mathbb{S}_A$, such that $\mathbb{S}_A \otimes_{\mathbb{S}_p} \mathbb{Z}_p \simeq A$ and the cyclotomic Frobenius of $\mathbb{S}_A$ is a lift of Frobenius on A. We let $A \to A_\infty$ be the p-complete colimit perfection.

Notations 3.2. We let S be a large quasisyntomic A-algebra. It becomes an $\mathbb{E}_\infty$-$\mathbb{S}_A$-algebra in a natural way.

We state the following lemma, which makes sure that the colimit perfection automatically admits an $\mathbb{E}_\infty$-spherical lift.

Lemma 3.3. *The p-complete colimit perfection A_∞ admits a unique lift to a p-complete connective $\mathbb{E}_\infty$-ring spectrum $\mathbb{S}_{A_\infty}$. The map $A \to A_\infty$ can be lifted to an $\mathbb{E}_\infty$-map $\mathbb{S}_A \to \mathbb{S}_{A_\infty}$.*

Proof. Since A_∞ is a perfect δ-ring, the lift $\mathbb{S}_A$ is uniquely given by the spherical Witt vector $\mathbb{S}_{W(A^\flat_\infty)}$ introduced in [[Lur17]; 5.2.7]. We now construct the map $\mathbb{S}_A \to \mathbb{S}_{A_\infty}$. Note that by the Segal conjecture for general objects (see for example [BS24]), $\mathbb{S}_A \simeq \mathbb{S}_A^{tC_p}$. Therefore, one can form the sequential colimit

$$\operatorname{colim}\left(\mathbb{S}_A \xrightarrow{\varphi_{tC_p}} \mathbb{S}_A^{tC_p} \simeq \mathbb{S}_A \xrightarrow{\varphi_{tC_p}} \mathbb{S}_A^{tC_p} \to \cdots\right)$$

By our requirement, φ_{tC_p} coincides with the Frobenius on π_0. Therefore, the above colimit yields an $\mathbb{E}_\infty$-lift of A_∞. We obtain the desired $\mathbb{E}_\infty$-map. □

We can see some examples of these spherical lifts.

Example 3.4. Any polynomial ring over $\mathbb{Z}_p$ admits a spherical lift. Indeed, we need only to check for $\mathbb{Z}_p[x]$, where the spherical lift is given by $\mathbb{S}_p[x] = \mathbb{S}_p[\mathbb{N}]$.

Example 3.5. Suppose A admits a spherical lift $\mathbb{S}_A$, then the polynomial ring $A[q]$ admits a spherical lift $\mathbb{S}_A[q]$. As a consequence, $A[\![q-1]\!]$ admits a spherical lift $\mathbb{S}_A[\![q-1]\!]$, where we write

$$\mathbb{S}_A[\![q-1]\!] \simeq \lim_\alpha \mathbb{S}_A[q]/(q^{p^\alpha}-1) \simeq \lim_\alpha \mathbb{S}_A[\mathbb{Z}/p^\alpha]$$

Moreover, one can check that the perfection of $A[\![q-1]\!]$ is isomorphic to $A_\infty[\![q^{1/p^\infty}-1]\!]$, with the spherical lift identified with $\mathbb{S}_{A_\infty}[\![q^{1/p^\infty}-1]\!]$

Now we can state our generalized version of the theorem.

Theorem 3.6. *We let A and S as in Notations 3.1 and Notations 3.2. Suppose additionally the canonical map*

$$\mathrm{THH}(\mathbb{S}_{A_\infty}) \to \mathbb{S}_{A_\infty}$$

is an equivalence. Then there is an equivalence of filtered rings.

$$\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S/\mathbb{S}_A)^\wedge_p \simeq \mathcal{N}^{\geqslant *}\hat{\mathbb{\Delta}}^{(1)}_{S/A}$$

Proof. We do everything in the p-complete category and simplify the notation of p-completion. We define $S_\infty := S \otimes_A A_\infty$. S_∞ is static and is of bounded p^∞-torsion as the Frobenius on A is p-completely faithfully flat. Also note that S is quasi-lci over A. Thus S_∞ is quasisyntomic.

By base change formula, we have

$$\mathrm{THH}(S/\mathbb{S}_A) \otimes_{\mathbb{S}_A} \mathbb{S}_{A_\infty} \simeq \mathrm{THH}(S \otimes_{\mathbb{S}_A} \mathbb{S}_{A_\infty}/\mathbb{S}_{A_\infty})$$

However, we require that

$$\mathrm{THH}(\mathbb{S}_{A_\infty}) \to \mathbb{S}_{A_\infty}$$

is an equivalence. Then we must have

$$\mathrm{THH}(S/\mathbb{S}_A) \otimes_{\mathbb{S}_A} \mathbb{S}_{A_\infty} \simeq \mathrm{THH}(S \otimes_{\mathbb{S}_A} \mathbb{S}_{A_\infty}) = \mathrm{THH}(S_\infty)$$

Therefore we have

$$\mathrm{THH}(S/\mathbb{S}_A) \otimes_A A_\infty \simeq \mathrm{THH}(S_\infty)$$

Now we apply homotopy fixed point functor $(-)^{hS^1}$ to both sides. Note by cyclotomic assumption, after taking homotopy fixed points, the left sife will add a Frobenius twist. Thus one obtains

$$\mathrm{TC}^-(S/\mathbb{S}_A) \otimes_\phi A_\infty \simeq \mathrm{TC}^-(S_\infty)$$

We then apply the double sheering graded pieces of HRW even filtration on both sides.

$$\left(\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S/\mathbb{S}_A)\right)\otimes_\phi A_\infty \simeq \Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S_\infty)$$

The right side is equivalent to $\mathcal{N}^{\geqslant *}\mathbb{\Delta}_{S_\infty} \simeq \mathcal{N}^{\geqslant *}\mathbb{\Delta}_S \otimes_A A_\infty$. Therefore, one obtains the desired equivalence by base change. □

Remark 3.7. If we apply the theorem to Example 3.4 and $S = \mathcal{O}_K$, then we will obtain [[BS22] ; Proposition 15.7].

3.2. **q-de Rham cohomology var Nygaard completion.** We can now apply the above consequence to the q-de Rham cohomology settings.

Theorem 3.8 ([Wag])**.** *Let S be a p-complete quasisyntomic $A[\zeta_p]$-algebra, then there is an equivalence of graded $\mathbb{E}_\infty$-$\mathbb{Z}_p[\![q-1]\!][u,v]/(uv-[p]_q)$-algebras*

$$\Sigma^{-2*}\mathrm{gr}^*_{\mathrm{HRW}}\mathrm{TC}^-(S/\mathbb{S}_A[\![q-1]\!])^\wedge_p \simeq \mathcal{N}^{\geqslant *}\hat{\mathbb{\Delta}}^{(p)}_{S/A[\![q-1]\!]}$$

Proof. Note that S becomes a quasisyntomic $A[\![q-1]\!]$-algebra by $q-1\mapsto\zeta_p$. By Example 3.5, $A[\![q-1]\!]$ admits the spherical lift $\mathbb{S}_A[\![q-1]\!]$ with the colimit perfection $\mathbb{S}_{A_\infty}[\![q^{1/p^\infty}-1]\!]$.

Suppose that S is large, then by Theorem 3.6, it suffices to prove that

$$\mathrm{THH}(\mathbb{S}_{A_\infty}[\![q^{1/p^\infty}-1]\!]) \to \mathbb{S}_{A_\infty}[\![q^{1/p^\infty}-1]\!]$$

is an equivalence. This can be argued as [[BMS19]; Proposition 11.7]. It suffices to prove

$$\mathrm{HH}(A_\infty[\![q^{1/p^\infty}-1]\!]) \to A_\infty[\![q^{1/p^\infty}-1]\!]$$

is an equivalence, then apply $-\otimes_{\mathbb{Z}}\mathbb{S}$ we obtain the desired equivalence. This reduces to the well-known result that $L_{A_\infty[\![q^{1/p^\infty}-1]\!]/\mathbb{Z}}$ is an equivalence.

In general, we can choose an extension of large objects $S\to A[x_i|i\in I]$(This can always be done by the remarks under Definition 2.14). Note that after apply THH, this map becomes an eff map of $\mathbb{E}_\infty$-rings, thus it satisfies the eff descent for the HRW-even filtration on the left side. It is also quasiregular semiperfectoid, thus it satiefies the quasisyntomic descent on the right side. Therefore, both sides coincide and we win. □

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